Handoff 05: assembly and bring-up

For: Matt at the bench, with Claude Code in VS Code following along. Claude reads the debug log, checks values against `docs/verification/VERIFY.md` and records them there. Allow about 3 h for the first lamp.

You need:

- **Tools:** fine-tip iron, hot plate and/or hot air, flux, IPA, tweezers, a 10× loupe, a multimeter with diode test, and a bench supply with a current limit (or a spare charged cell plus the meter).
- **Stencils and jig:** 0.12 mm stencils, plus the printed `jig_led_board` (PETG, from `print/`).
- **Fitment views:** `docs/design/fitment/`, rendered by `cad/tools/fitment.py`. They show where every part sits: the exploded pack and moon head, and sections through the pack and the head.
- **Drawings:** `fab/<board>/v0.1/assembly_top.pdf` / `assembly_bottom.pdf`. These have values on the Fab layer and J4 crossed out as DNP. The zoomed views are easier to read: `fab/<board>/v0.1/renders/*_assembly_*.png`.
- **Off-board items** (`bom/other_purchases.csv`): Kapton tape, PTFE sleeve, heat-shrink, CA gel, and neutral-cure silicone (or E6000).

1. Main board (all parts on top)

1. Inspect the bare board. There must be no shorts between VBUS / BAT_RAW / VSYS / VLED / GND: meter TP1, TP2, TP3 and TP7 to TP4 (GND).

2. Stencil and paste, then place the parts per the assembly drawing:

   - the passives;
   - SOT-23s Q1–Q4, checking the orientation dots;
   - U1 MCP73831 and U2 ATtiny3216;
   - D1/D3 with the cathode bar to the silk bar, and the red 0603 D2 with its cathode mark to pad 1;
   - J1 USB-C, SW2–4 tact switches, J2 PH and J3 SH.

   Reflow (SAC305, or leaded paste for a lower peak). Then solder J1's four shell slots through the holes with the iron.

3. **Fit SW1 (PCM12) by iron after the reflow.** It is rated 255 °C max, so it must not go through a hot plate (round2 R2-D13). **Do not fit J4**: hold a header in the holes when programming. A soldered header would point at the pouch (round1 B-2).

4. Clean off the flux. Under the loupe, check U2's pins and the USB-C pads for bridges.

5. Run VERIFY **E1–E4** now (bench supply on J2, no cell, no LED board). E2 finds which knob direction is ON: mark it.

6. Flash the firmware: handoff 04, steps 1–5. With no LED board fitted, the lamp just retries quietly. The debug log shows `T=-99` and then the 30 % fail-safe cap, which is expected.

2. LED board (bottom first, then top)

1. **Bottom side** (its own stencil): U1 TLC59116 (TSSOP-28, pin-1 dot), U2 LM75BDP (TSSOP-8), R1 750 Ω, R2 2.4 kΩ, C1–C3. Reflow on the hot plate, then check every TSSOP pin for bridges under the loupe. Put a **Kapton disc over U1** afterwards: it faces the gooseneck's metal tube end (round2 R2-D3).

2. **Top side.** Put the board in `jig_led_board`, LEDs up. The jig supports only the part-free edge band, so the bottom parts hang free and the board top is flush with the jig top. Stencil, then place:

   - **8 × white 2835.** The silk letter next to each one says which: **W = 2700 K** (D1, D3, D5, D7) and **C = 5000 K** (D2, D4, D6, D8). The cathode is the large pad and goes to the silk cathode mark.
   - **2 × Würth RGB (D9, D10).** The package's polarity mark (chamfer/dot) is **pin 3, the red cathode**: put it on the **silk chamfer**. Pin 1, the common anode, then lands on the pad marked "1". Don't put the package's mark on the "1" (round2 R2-D4).

   Reflow with hot air, ≤ 260 °C and ≤ 10 s at peak. Or lift the board off the jig onto two metal spacers under the edge band and use the hot plate. The jig is PETG, so it is for stencilling and hot air, not a 240 °C plate. The 2835s have bottom-only terminations, so an iron can't solder them. The bottom parts will re-melt; surface tension holds them.

3. Diode-test each LED: + probe on pad V (VLED, all anodes), − probe on its TLC output pin. Each one should glow faintly in the right colour; a dark one is reversed or open. Green and blue need about 2.6 V to glow, so a meter whose diode test gives less can only check them at first light.

3. Cable and neck (round1 M-7: abrasion and strain relief)

Qwiic cable colours: **black = GND, red = VLED, blue = SDA, yellow = SCL**. The LED board pads are labelled **G V D C** on its bottom silk.

1. Deburr the inside of both gooseneck ends with a needle file. Self-tap the gooseneck's lower end into `pack_front`'s 9.3 mm socket by hand, before any cable is fitted, so nothing twists. A drop of CA gel on the thread is optional. **No anaerobic thread-locker**: it crazes PETG (round1 m-7).
2. Cut the Qwiic cable to about 200 mm, keeping the JST-SH end. Slide the PTFE sleeve over the 4 wires. Feed the bare end up through the gooseneck from the pack side. Put 15 mm of heat-shrink over the wires where they leave each threaded end.
3. Push the gooseneck's top end up through the `moon_base` floor. **The thread must not stand more than 0.5 mm above the nut top**: file it flush if it does (VERIFY M12). Thread the wires through the M10×1 nut, then spin the **nut** (not the gooseneck) down into the hex socket. The wires leave the bore above the nut, into the 4.5 mm space under the board.
4. Solder G / V / D / C on the LED board. Cover the joints and 5 mm of wire with neutral-cure silicone, and keep the blob **inside r ≈ 11 mm**, toward the board centre: silicone on the ledge band would tilt the board. Let it cure. This is the strain relief.
5. Lay the LED board in the base pocket (LEDs up; it stands 0.12 mm proud of the step), then put the PETG `moon_ring` on its edge band. Fit the moon: lugs into the entry slots, then turn 30° to lock. To remove the moon, **hold the base cup**, so the turn doesn't unscrew the base from the neck.

4. Pack

1. Put Kapton tape over the top face of the pouch, and a patch on the PCB back under U1, D3 and Q1–Q3.
2. **VERIFY E1 again on the real cell: + must be on J2 pin 1.** From the back of the PCB, pass the cell's plug up through the 6.4 × 5.0 mm slot beside J2, mate it with J2, and let it lie flat over the slot. Lay the slack lead flat in the bay's end gap, clear of the two screw heads (VERIFY M13).
3. Fit the button caps into the front shell. Plug the JST-SH into J3. Leave a 10–15 mm service loop, then fix the cable to the gooseneck bore entrance in the neck block with a dab of silicone. A pull on the neck is then taken there and never reaches the JST-SH, which has no latch. Screw the PCB onto the bosses with 2 × M2×6.
4. Battery into the bay. Hook the back-plate tabs into the slots, then fit 2 × M2×12 through the clip root and the plate into the front shell.

5. First light and checks

1. Switch ON with no USB. The lamp should fade into Soft white 3000 K at level 10. Run through the modes, UP/DOWN and hold-to-save (two dips).
2. Debug build (handoff 04 §5): run VERIFY **F2–F5** and **E7–E15**. E13 is the scope check of the LED-board soft turn-on.
3. Before regular use, run at least **T3** (4000 K full for 2 h with thermocouples, or a contact thermometer on the moon skirt) and **T6** (the thermostat). Record the results in VERIFY.md and send them to Claude, who re-fits the derating if the board-to-PLA offset is not about 14 K.
4. Light and colour: VERIFY **O1–O7** as equipment allows. A phone lux app is good enough for O3.

6. User notes (for whoever uses the lamp)

- Charge at room temperature (the charger has no cell-temperature sensing). The red LED means charging. It flickers when no battery is connected, which is normal.
- The side switch is the only off. Switch off when you put the lamp down.
- After a "battery empty" fade-out, plug in USB. The lamp comes back on by itself after a few seconds.
- If the pack ever bulges, stop using it and replace the cell.
- Never power the lamp from a programmer or a USB-UART adapter.

Done when

One lamp is assembled and every E, F and T3/T6 item has a measured value in VERIFY.md. Anything outside its pass band is raised with Claude before the second lamp is built.